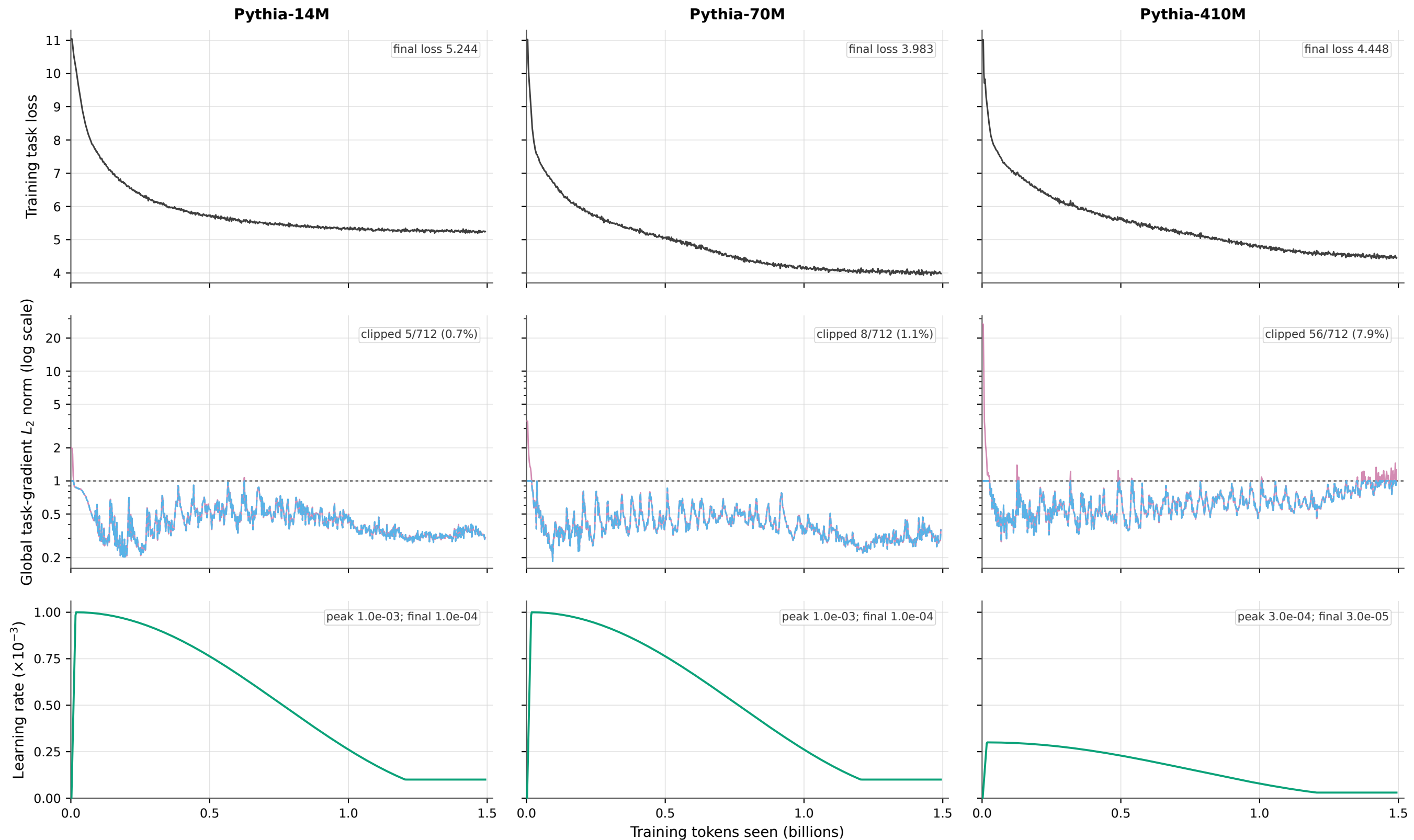


A0 optimization trajectories through one MiniPile pass

Training task loss, global full-model L2 gradient norm, and learning rate at every optimizer boundary

— Training task loss — Learning rate — Before clipping - - - After clipping - - - Global clip threshold = 1



Raw boundary values; no smoothing. Each boundary contains 2,097,152 tokens. All 712 updates completed without overflow or skipping.
Learning rates use a shared absolute scale. The gradient norm is not parameter-count normalized.